Execution Procedure

Team Oriented Project

Group A

Linux Kernel Extension for DM-Verity Authentication, Verification and Setup

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After successfully installing the packages mentioned in the *Technical Documentation - Prerequisites.pdf*, you are ready to proceed with these execution steps.

Execution Procedure 1: Running with the Original Root Filesystem

This procedure covers building the system artifacts and launching the authenticated system environment using QEMU.

- **1. Navigate to the build directory:** Change your current directory to **`/build`**.
- **2. Create the artifacts:**
    - Make the build script executable: `chmod +x build_artifacts.sh`
    - Run the script: `./build_artifacts.sh`
- **3. Start QEMU:**
    - Make the QEMU launch script executable: `chmod +x launch_qemu.sh`
    - Run the script: `./launch_qemu.sh`
    - Choose Option 1 to boot from original rootfs image
- **4. Log In:** Once the system has booted, log in using the credentials: Username: root, Password: root.
- **5. Verify dm-verity inside the rootfs:** Execute the command `dmesg | grep -i verity`.
- **Verification Result:** You should now be able to see that the Kernel verifies the signature, parses the metadata, and created the device mapper successfully.

Execution Procedure 2: Running with the Docker Environment

This procedure outlines using a pre-configured Docker environment for the DM-Verity setup.

- **1. Navigate to the source directory:** Change your current directory to **`/src`**.
- **2. Set up the Docker environment:**
    - Make the setup script executable: `chmod +x docker_setup.sh`
    - Run the script: `./docker_setup.sh` (**A system reboot is expected after this step**).
- **3. Start the Docker container:**
    - Make the run script executable: `chmod +x docker_run.sh`
    - Run the script: `./docker_run.sh`
- **4. Choose Option:** Select option **1** to run the original rootfs image.
- **5. Log In:** Log in to the container's rootfs using the credentials: Username: root, Password: root.

- **6. Verify dm-verity inside the container rootfs:** Execute the command `dmesg | grep -i verity`.
- **Verification Result:** You should now be able to see that the Kernel verifies the signature, parses the metadata, and created the device mapper successfully.

Testing Procedures

These procedures cover running the automated test suite and individual integrity tests.

A. Running the Test Wrapper

- **1. Navigate to the build directory:** Change your current directory to **`/build`**.
- **2. Run the test wrapper:**
    - Make the script executable: `chmod +x run_verity_tests.sh`
    - Run the script: `./run_verity_tests.sh`

B. Running a Specific Integrity Test

- **1. Navigate to the build directory:** Change your current directory to **`/build`**.
- **2. Run the integrity tests script:**
    - Make the script executable: `chmod +x integrity_tests.sh`
    - Run the script: `./integrity_tests.sh`
- **3. Run a specific integrity test:** Execute the script, providing the test name as an argument: `./integrity_tests.sh <test_name>`
  (use command `./integrity_tests.sh help` to see available tests names.)
- **4. Launch QEMU:**
    - Make the QEMU launch script executable: `chmod +x launch_qemu.sh`
    - Run the script: `./launch_qemu.sh`
- **5. Select the corrupted image:** When prompted by QEMU, select the corrupted image that you created before.
- **Expected Outcome:** Depending on the test, the kernel may panic, reject boot, or load the rootfs normally based on the DM-Verity verification status.

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